

Task 3: Avatar Coaching Assistant (Webapp)

 **Date** 

Jul 6, 2025

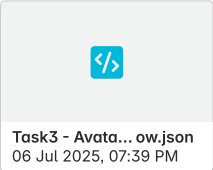
 **Participants** 

- @TASNIM MAHDIYA / UPM

 **[Task 3 documentation link](#)** 

 [Audio to convert speech to text \(test\)](#)

 **n8n workflow** 



 **Project Overview** 

This project implements a **Conversational Avatar Coaching Assistant**, simulating a human-like coach that engages with users via a voice-based chat interface. The assistant receives user voice input, processes it with an LLM, generates a coaching response, creates an avatar video, and is meant to deliver the video to a user-facing UI.

 **Tech Stack** 

 **Frontend Stack:** 

Tech	Purpose
React + MUI	Responsive web interface for viewing avatar
Axios	API call to backend <code>n8n</code> to fetch video URL
Vite	Build tool for fast dev environment

 **Workflow Architecture** 

```
1 User Input (Webhook Trigger)
2   ↳ LLM - DeepSeek
3     ↳ Code (Parse JSON & Format Date)
4       ↳ Generate Avatar Video (HeyGen)
5         ↳ Store Video ID + Status
6           ↳ LOOP: Get Video Status (Wait every 20s, max X loops)
7             ↳ [IF loop count ended || video Completed?]
```

```
8      |↳ Record to Airtable (both)
9      └─[IF Completed?]
10     |↳ True: Use Generated URL
11     └─False: Use Fallback Video URL
12         ↳ Respond to Webhook (return final URL)
```

Backend Logic (n8n) [🔗](#)

1. Webhook Trigger

- Method: POST
- Accepts user input to begin avatar coaching session

2. DeepSeek LLM [🔗](#)

- Prompt: Coaching assistant behavior
- JSON Output containing reply text

3. Generate Avatar Video (HeyGen API) [🔗](#)

```

1 POST
2 Headers:
3   X-API-Key: <API_KEY>
4 Body:
5 {
6   "video_inputs": [
7     {
8       "character": {
9         "type": "avatar",
10        "avatar_id": "Lina_Dress_Sitting_Side_public",
11        "avatar_style": "normal"
12      },
13      "voice": {
14        "type": "text",
15        "input_text": "{{ $node['LLM - Coach
Reply'].json.choices[0].message.content.replace(/\\n/g, ' ').replace(/\\\"/g, '\\\\\"') }}"
16        "voice_id": "119caed25533477ba63822d5d1552d25",
17        "speed": 1
18      }
19    }
20  ],
21  "dimension": {
22    "width": 1280,
23    "height": 720
24  },
25  "test": true
26 }
27

```

4. Loop: Check Video Status [🔗](#)

- GET from: `$json.data.id }}`
- Wait Node: 20s between loops
- Loop Max: 30

- No authentication required for internal test

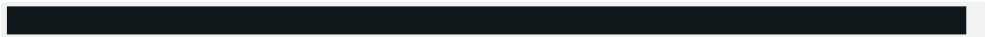
 Limitations [🔗](#)

Area	Limitation / Workaround
 LLM	DeepSeek can parse and reply properly, but may return date as "2024" which was manually fixed to "2025" in the code
 Avatar	HeyGen Free Tier allows 5 runs/day. After that: "AVATAR_USAGE_NOT_PERMITTED" error
 Looping	<code>loop_count</code> logic breaks due to re-execution context loss — stays at 1
 Video Processing	Can take up to 2–3 minutes or longer; often exceeds real-time demo constraints
 Video URL	Sometimes HeyGen dashboard doesn't show the video even when generation succeeds — URL must be extracted manually
 Webhook	Workflow Webhook node is not properly connected to frontend — real-time user interaction is not available in final build

 Sample Fallback URL Used [🔗](#)

```
1 arduino
```

CopyEdit



Shown when video generation fails or times out.

 Sample Output Sent to Airtable [🔗](#)

Field	Value (Example)
ID	80bee38ea8944f37967dbce7fabf5e24
Audio Transcript	A king ruled the state in early days...
AI Reply	It sounds like you're sharing vivid imagery...

Video URL	HeyGen generated Video URL...
Status	completed
Notes	Loop #1 (fallback used or not)

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CoachingSessions ▾ + Add or import										Top
Grid view ▾							Hide fields Filter Group IT Sort Color IT Share and sync			
	A ID	Timestamp	Audio Transcript	AI Reply	Video-URL	Status	Notes			
1	80bee38ea8944f379...	7/5/2025 1:36pm GMT	=A king ruled the state in early days. The...	It sounds like you're sharing some vivid imag...	https://files2.heygen.ai/ews_pacific/avatar_tmp/b2d35ac8...	completed	Loop #1			

Sample proof in Airtable

✔ Features Highlight 🔗

- Full coaching assistant experience
- Humanlike avatar using HeyGen
- LLM-integrated dynamic response
- Multilingual support (Malay + English)
- Airtable logging for all sessions
- Seamless UI integration
- Robust fallback logic with loops